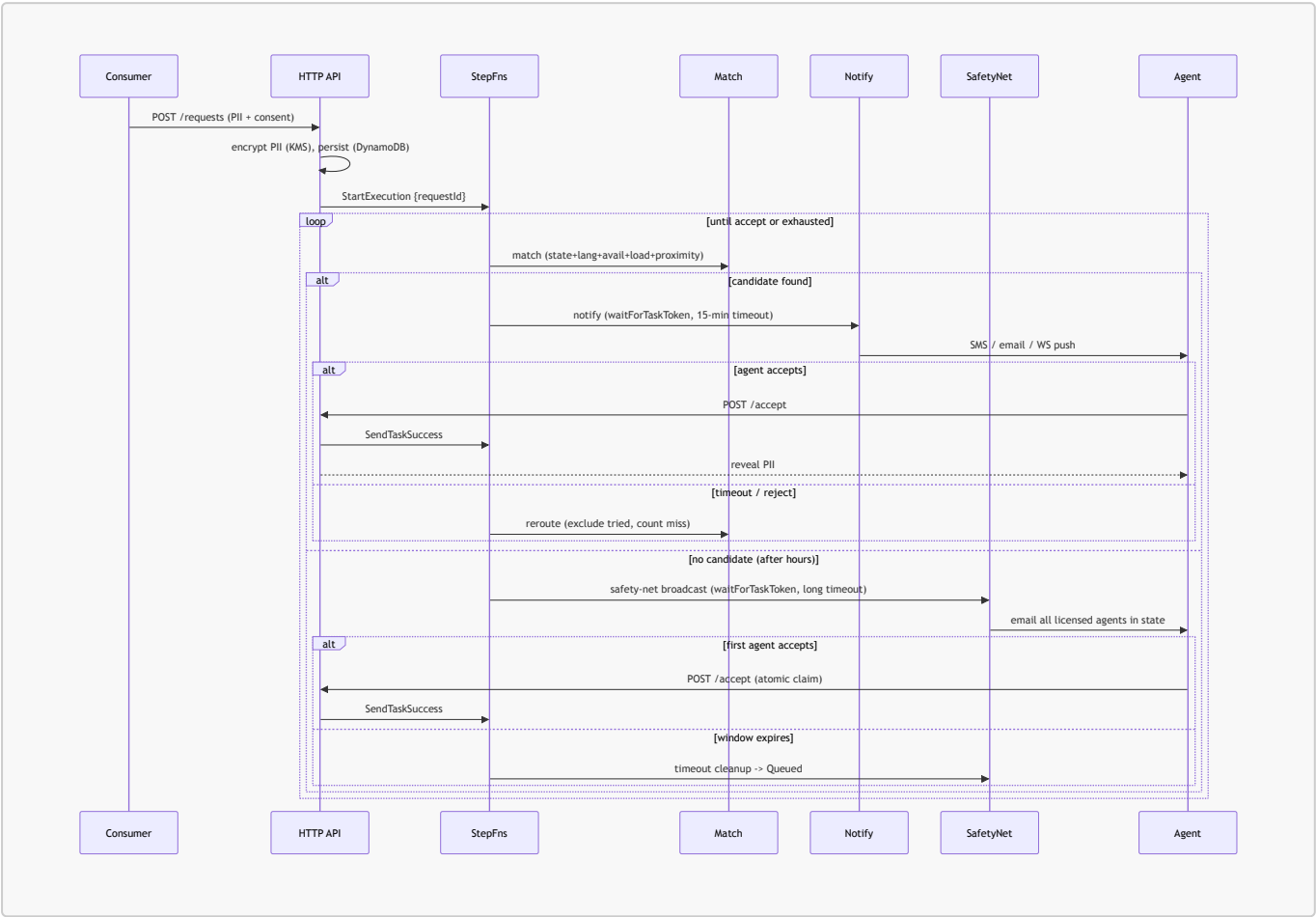


RAN Architecture

Components

Layer	Technology	Responsibility
Frontend	React + TypeScript (Vite)	Consumer intake, agent dashboard, admin dashboard
API (REST)	API Gateway HTTP API → Lambda	Request submission, agent actions, admin queries
API (real-time)	API Gateway WebSocket → Lambda	Live status to consumer/agent/admin
Compute	Lambda (Node/TS)	Stateless business logic
Orchestration	Step Functions	Per-request routing + 15-minute reroute timer
Data	DynamoDB	Agents, requests, connections, config
Security	KMS, Cognito	PII encryption, agent/admin identity & RBAC
Notifications	SNS, SES	SMS + email to agents/brokers

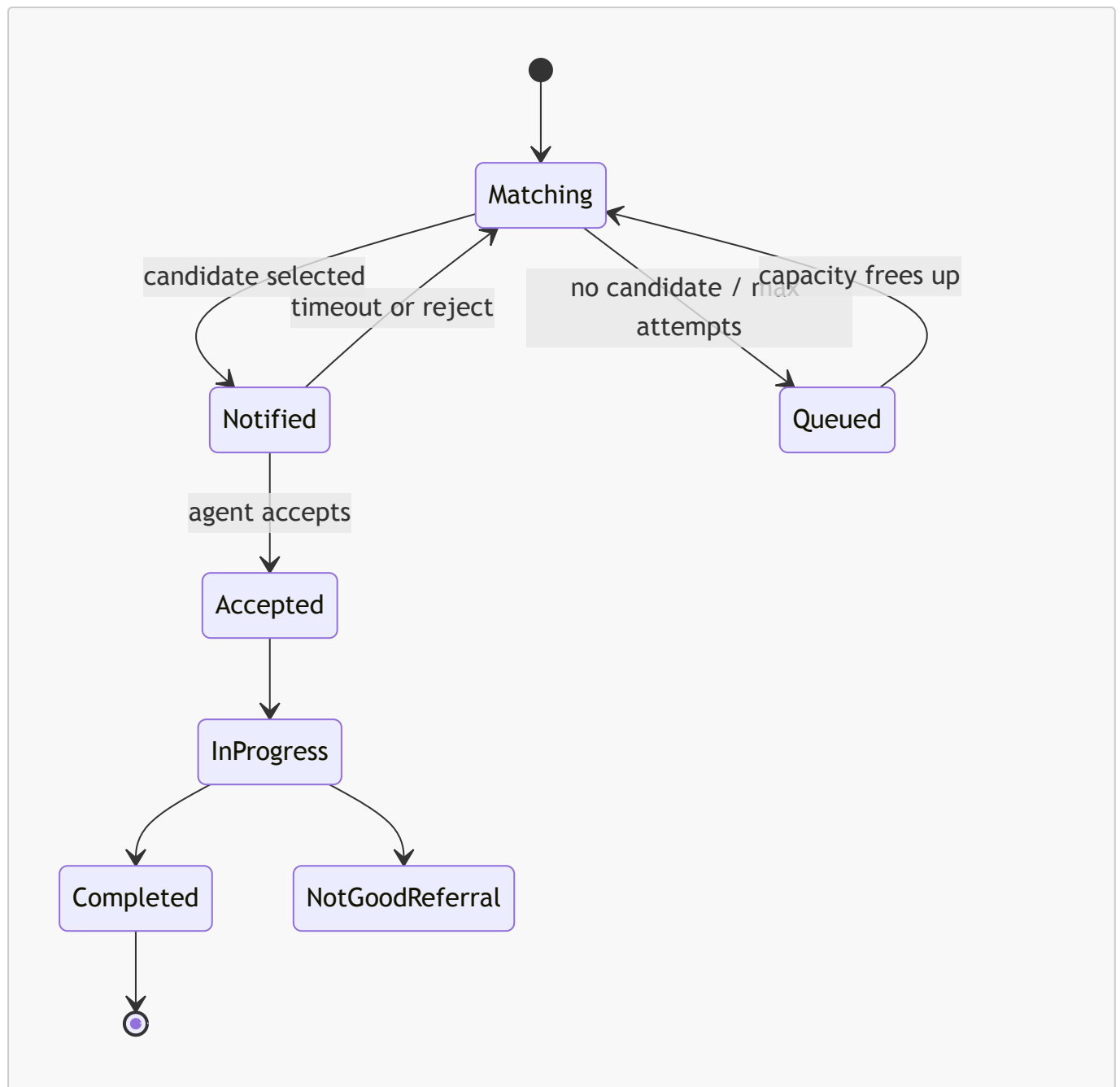
Request lifecycle



After-Hours Consumer Safety Net

When no immediately-available agent is found (all offline / outside standard hours / at capacity), the request falls through to the **SafetyNet** state: `sfnSafetyNet` emails every licensed, participating, training-current agent in the consumer's state (ignoring online status and hours). This is **email-only** and carries **no penalty**, matching HOD policy. The first agent to accept wins via an atomic DynamoDB conditional claim (`claimSafetyNetRequest`); other acceptors get a 409 and cannot access the consumer's PII. If the safety-net window expires with no acceptance, `sfnSafetyNetTimeout` marks the request **Queued**.

Request status state machine



Matching algorithm

Eligibility filter, then ranking (lower score is better):

- **Eligibility:** agent online, annual training current, not Out of Office, ZIP→state in **activeStates**, language supported, **currentLoad** < **maxLoad**, current time within an availability window (or within a "Today's Availability" override), not already tried for this request.
- **Ranking:** $\text{loadFactor} \cdot (1 - w) + \text{proximityFactor} \cdot w + \text{recencyFactor} \cdot 0.1$ where **w** is the admin-configurable **proximityWeight**. Load balancing dominates (per the SOO objective) so referrals spread evenly across agents in a geography; proximity (by zip-code center) and least-recently-assigned break ties. This generalizes the current HOD behavior, which routes to the closest agent by center of zip code.

Exclusions are derived from each request's **routingHistory**, so the state machine itself stays stateless about which agents were already attempted.

Tables

- **ran-agents** (PK **npi**, GSI **byState**) — profile, licensure, availability, live load.
- **ran-requests** (PK **requestId**, GSI **byStatus**) — lifecycle + encrypted PII + task token.
- **ran-connections** (PK **connectionId**, GSI **byChannel**, TTL) — WebSocket registry.
- **ran-config** (PK **configKey**) — admin-editable runtime rules.